

BLEVE Peak Overpressure Prediction

Machine Learning Regression Project

Tech Stack:

Python • Scikit-Learn • XGBoost • Neural Networks • Data Preprocessing • Feature Engineering

Models Implemented:

ElasticNet | Support Vector Regression | XGBoost | Deep Neural Network Ensemble

Key Result:

Achieved 0.17849 MAPE using an optimized XGBoost + Neural Network ensemble approach

Introduction

Boiling Liquid Expanding Vapour Explosions (BLEVEs) represent one of the most severe industrial hazards associated with the transportation and storage of liquefied gases such as LPG. A BLEVE occurs when a pressurised vessel fails catastrophically, causing the superheated liquid inside to instantaneously vaporise and generate a powerful blast wave. The resulting peak overpressure determines the severity of structural damage and risk to human life in surrounding areas.

Predicting peak overpressure accurately is challenging due to its dependence on a complex, non-linear combination of physical factors — including tank geometry, substance thermodynamic state, explosion height, and sensor position relative to nearby structures. Traditional physics-based simulation methods are computationally expensive and difficult to generalise across scenarios. Machine learning offers a data-driven alternative capable of learning these relationships directly from simulation data.

This report documents the complete machine learning pipeline developed. The objective is predicting peak pressure at 27 sensor locations around a rigid obstacle wall, given 23 input features describing each BLEVE scenario. The training dataset comprises 10,050 observations across approximately 370 unique scenarios. Four fundamentally different model types were evaluated — ElasticNet, Support Vector Regression, XGBoost, and a Neural Network — before combining the two strongest models into a weighted ensemble. The final model achieved a Kaggle public leaderboard MAPE of 0.17849.

1. Data Cleaning

1.1 Missing values

The raw training set (10,050 rows, 25 columns) had missing values in every column, ranging from 5 to 30 per column. Median imputation was applied to all numeric columns (fit on train only, applied to test — no leakage). Rows with missing target values (10 rows) were dropped. Medians were preferred over means due to the heavy right-skew of the target distribution (mean 0.360 bar, max 9.17 bar).

1.2 Duplicate Rows

Exactly 50 exact duplicate rows were identified in the training data using pandas `drop_duplicates()`. These were removed before model training, as keeping duplicates would artificially increase the weight of those scenarios and bias the model.

1.3 Status Column Typos

The Status column contained 9 distinct variants due to inconsistent data entry. All variants were standardised to a binary encoding (1 = Superheated, 0 = Subcooled) using the mapping below.

Raw Value	Correct Label	Fix
Superheated / Superheat / superheated / Saperheated	Superheated	Map → 1
Subcooled / Subcool / subcooled / Subcoled	Subcooled	Map → 0
NaN (7 rows)	Physics-inferred	Vapour T > Liquid T → 1, else → 0

1.4 Outlier Removal

Two outlier categories were removed from the training data only:

- **Extreme Tank Failure Pressure (>50 bar):-** 50 training rows had pressures up to 4,995 bar. The test set maximum is only 42 bar. These rows teach the model patterns irrelevant to the test distribution and were removed.
- **Extreme target outliers (>Q99.9):-** 10 rows with implausibly high peak pressures (simulation artefacts) were removed using a 99.9th percentile threshold.

2.Data Processing

2.1 Feature Engineering

23 physics-informed features were derived from the raw variables to better capture the underlying relationships governing blast wave propagation.

The complete set of 23 engineered features is: Tank_Volume, Width_Length_Ratio, Height_Length_Ratio, Vapour_Height_Ratio, Vapour_Volume, BLEVE_Height_Ratio, Tank_Energy, Pressure_per_Volume, Pressure_LiquidRatio, Pressure_VapTemp, Critical_Ratio, log_Pressure, log_Distance, Temp_Diff, Superheat_Degree, Obstacle_Volume, Obstacle_H_W_Ratio, Distance_to_Sensor, Sensor_z_norm, Sensor_x_norm, Sensor_y_abs, Sensor_Angle, Scaled_Pressure.

Feature	Formula	Physical Rationale
Tank_Volume	$\text{Width} \times \text{Length} \times \text{Height}$	Total gas volume — scales with blast energy released
Width_Length_Ratio	$\text{Width} / \text{Length}$	Tank shape affects blast directionality
Height_Length_Ratio	$\text{Height} / \text{Length}$	Aspect ratio of explosion source geometry
Vapour_Height_Ratio	$\text{Vapour H} / \text{Tank H}$	Fraction of tank occupied by vapour phase
Vapour_Volume	$\text{Width} \times \text{Length} \times \text{Vapour H}$	Volume of vapour available for expansion
BLEVE_Height_Ratio	$\text{BLEVE H} / \text{Tank H}$	Ground clearance relative to tank size
Tank_Energy	$\text{Pressure} \times \text{Tank Volume}$	Primary stored energy — drives overpressure
Pressure_per_Volume	$\text{Pressure} / \text{Tank Volume}$	Energy density — pressure concentration
Pressure_LiquidRatio	$\text{Pressure} \times \text{Liquid Ratio}$	Interaction of pressure and liquid fraction
Pressure_VapTemp	$\text{Pressure} \times \text{Vapour Temp}$	Combined thermodynamic energy proxy
Critical_Ratio	$\text{Pressure} / \text{Critical Pressure}$	Proximity to critical point — governs flashing
log_Pressure	$\log(1 + \text{Pressure})$	Log-linearises wide tank pressure range
log_Distance	$\log(1 + \text{Distance})$	Distance-decay of blast wave follows log scale
Temp_Diff	$\text{Vapour T} - \text{Liquid T}$	Degree of superheat — drives explosive vaporisation
Superheat_Degree	$\text{Vapour T} - \text{Boiling T}$	How far above atmospheric boiling point
Obstacle_Volume	$\text{W} \times \text{H} \times \text{Thickness}$	Obstacle mass — affects blast reflection pattern
Obstacle_H_W_Ratio	$\text{Height} / \text{Width}$	Aspect ratio of obstacle wall
Distance_to_Sensor	$\sqrt{x^2 + y^2 + z^2}$	True 3D Euclidean distance from explosion source
Sensor_z_norm	$z / \text{Obstacle Height}$	Relative sensor height on wall (0=bottom, 1=top)
Sensor_x_norm	$x / \text{Obstacle Distance}$	Normalised horizontal sensor position
Sensor_y_abs	$ y $	Lateral displacement from centreline
Sensor_Angle	$\arctan2(y, x)$	Azimuthal angle of sensor relative to blast source
Scaled_Pressure	Derived composite	Scaled overpressure — key pressure-distance feature

Feature importance analysis (Section 5.3) confirmed that engineered features dominate the top predictors — Scaled_Pressure, Vapour_Height_Ratio, Distance_to_Sensor, and Tank_Energy all rank in the top 10, validating the feature engineering choices. Raw features alone would not capture these composite physical relationships.

2.2 Categorical Encoding

Sensor Position Side (nominal, walls 1–5) was encoded using One-Hot Encoding. The encoder was fit on combined train+test data to handle all categories present in the test set. Label encoding was rejected as there is no ordinal relationship between wall sides.

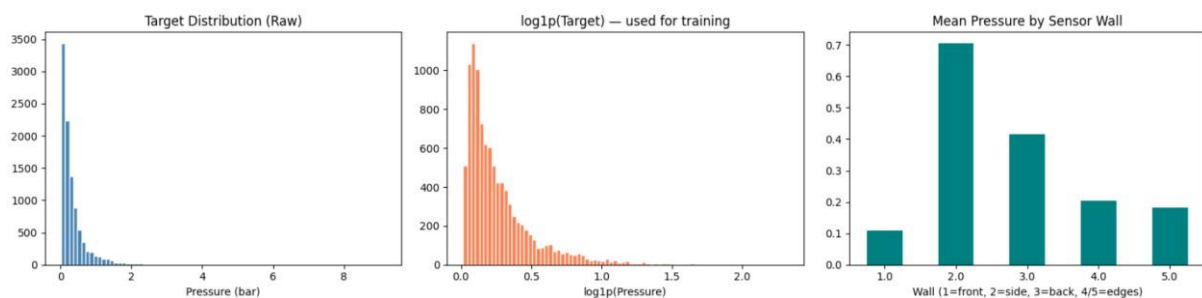
2.3 Feature Scaling

StandardScaler (zero mean, unit variance) was applied for ElasticNet, SVR, and the Neural Network, all of which are sensitive to input magnitude. XGBoost is scale-invariant and was trained on unscaled features. Scalers were fit only on training folds during cross-validation.

2.4 Target Log-Transformation

All models were trained on $\log_{10}(\text{target})$ and predictions were back-transformed using expm1 . This normalises the heavily right-skewed target distribution, reduces the influence of high-pressure outliers, and effectively optimises for relative errors — directly aligned with the MAPE evaluation metric.

Prior to transformation, the target distribution was heavily right-skewed (mean = 0.360 bar, std = 0.495, max = 9.17 bar, with 75% of values below 0.413 bar). After $\log_{10}$ transformation, the distribution became approximately symmetric, reducing the dominance of high-pressure scenarios during training and aligning the loss function with percentage-based evaluation.

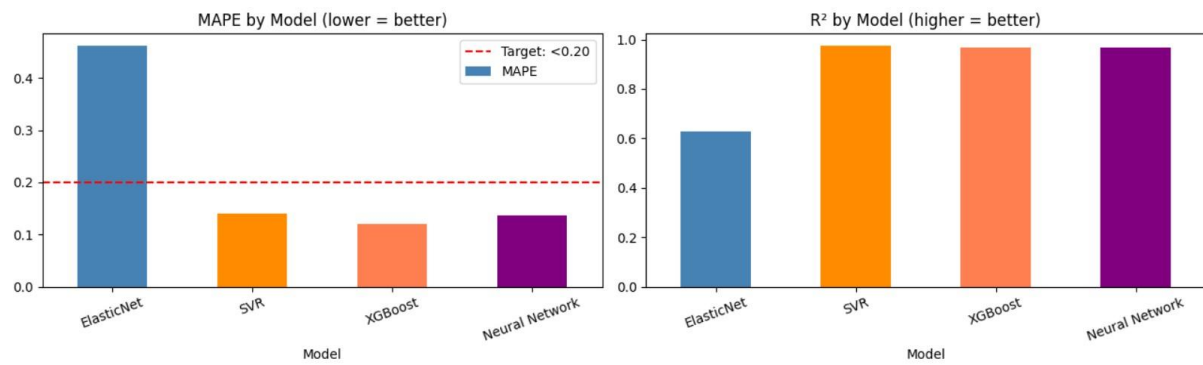


Left — raw target distribution showing heavy right-skew. Centre — $\log_{10}(\text{target})$ used for training, approximately symmetric. Right — mean pressure by sensor wall, confirming wall 2 (side) receives ~6x higher pressure than wall 1 (front).

3. Model Selection

Four fundamentally different model types were evaluated, selected to cover the spectrum from interpretable linear models to complex non-linear approaches.

Model	Type	Core Mechanism	CV MAPE	Role
ElasticNet	Linear	L1+L2 regularised weighted sum	0.4620	Baseline
SVR	Kernel	RBF kernel + epsilon margin	0.1396	Evaluated
XGBoost	Tree ensemble	Sequential residual boosting	0.1199	Final ensemble
Neural Network	Deep learning	Layered non-linear transformations	0.1373	Final ensemble



Cross-validation MAPE (left) and R^2 (right) for all four models. The red dashed line marks the 0.20 MAPE target. XGBoost and Neural Network both fall well below it.

-**ElasticNet** serves as the interpretable linear baseline. Its poor MAPE of 0.4620 confirms that blast overpressure relationships are strongly non-linear and cannot be adequately captured by a linear model.

-**SVR** with an RBF kernel implicitly maps inputs into a high-dimensional feature space, enabling non-linear regression. It optimises a margin-based objective fundamentally different from both linear and tree-based approaches, achieving CV MAPE of 0.1396.

SVR was ultimately excluded from the final ensemble. Although it achieved strong individual performance (CV MAPE 0.1396), its out-of-fold predictions were highly correlated with XGBoost, meaning combining them offered negligible diversity benefit. The XGBoost and Neural Network pairing was selected instead due to their structurally different error patterns.

-**XGBoost** builds trees sequentially, each correcting the residuals of the previous. It naturally handles feature interactions and non-monotone relationships, achieving the best individual CV MAPE of 0.1199 with early stopping to prevent overfitting.

-**Neural Network** learns hierarchical representations through stacked non-linear transformations with BatchNorm, GELU activations, and Dropout regularisation. Its complementary prediction errors relative to XGBoost make it valuable for ensembling, achieving CV MAPE of 0.1373.

4. Hyperparameter Tuning

ElasticNet, SVR, and XGBoost were tuned using RandomizedSearchCV with 3-fold CV. RandomizedSearch was chosen over GridSearch for efficiency — it samples random combinations rather than exhaustively evaluating all, making it tractable for models with many parameters.

ElasticNet — 15 iterations × 3 folds = 45 fits

Hyperparameter	Search Range	Best Value	Description
alpha	[0.0001 – 1.0]	0.05	Overall regularisation strength
l1_ratio	[0.1 – 1.0]	0.3	L1 vs L2 balance (0=Ridge, 1=Lasso)

SVR — 15 iterations × 3 folds = 45 fits

Hyperparameter	Search Range	Best Value	Description
C	[10, 50, 100, 200, 500]	10	Penalty for margin violations
gamma	['scale', 'auto', 0.01, 0.05, 0.1]	scale	RBF kernel width
epsilon	[0.01, 0.05, 0.1, 0.2]	0.01	Epsilon-insensitive tube width

XGBoost — 25 iterations × 3 folds = 75 fits

Hyperparameter	Search Range	Best Value
n_estimators	[500, 800, 1000, 1200]	800
max_depth	[4, 5, 6, 7]	5
learning_rate	[0.01, 0.03, 0.05, 0.08, 0.1]	0.05

subsample	[0.7, 0.8, 0.9]	0.9
colsample_bytree	[0.7, 0.8, 0.9]	0.9
min_child_weight	[1, 3, 5]	3
reg_alpha	[0, 0.01, 0.1, 0.5]	0.1
reg_lambda	[1, 1.5, 2]	2

Early stopping (50 rounds on validation fold) was applied during 5-fold CV training.

Neural Network — Validated on held-out fold

Hyperparameter	Value	Rationale
Architecture	512 → 256 → 128 → 64 → 1	Progressive narrowing — hierarchical representation
Activation	GELU	Smooth gradients — better than ReLU for regression
Dropout	0.2	Prevents co-adaptation of neurons
Optimiser	AdamW (lr=3e-4, wd=1e-4)	Decoupled weight decay for better regularisation
LR Schedule	CosineAnnealingLR (T=300)	Smooth decay — avoids sharp learning rate drops
Loss	HuberLoss (delta=0.5)	Robust to large residuals compared to MSE
Early stopping	Patience = 40 epochs	Terminates training when validation MAPE stalls

Parameters validated via manual search on a held-out validation fold. The ranges above were explored before settling on final values.

Architecture depth (3 vs 4 vs 5 layers), dropout rates (0.1, 0.2, 0.3), and learning rates

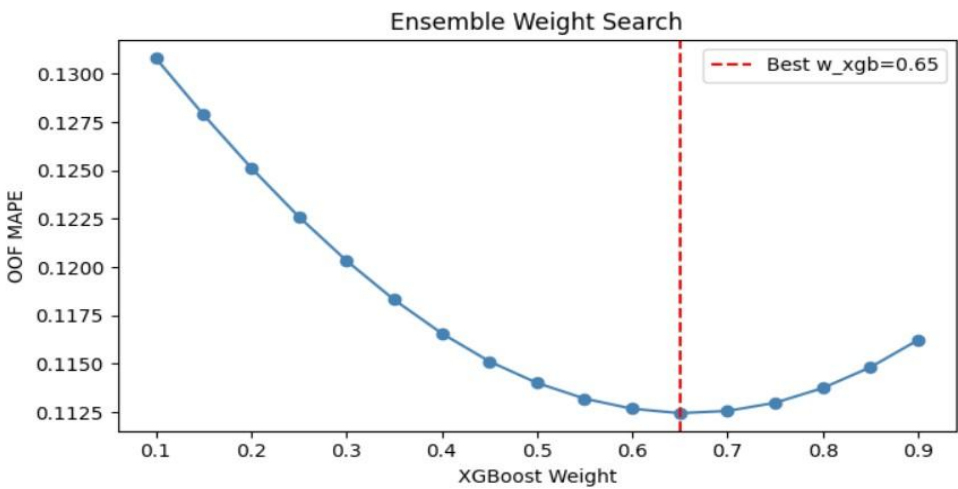
(1e-3, 3e-4, 1e-4) were each tested on a held-out fold before the above configuration was finalised.

5.Prediction Performance

5.1 Cross-Validation Results (5-Fold KFold, SEED=42)

Model	MAPE	R ²	RMSE	MAE
ElasticNet	0.4620	0.6274	0.2792	0.1328
SVR	0.1396	0.9752	0.0718	0.0346
XGBoost	0.1199	0.9678	0.0820	0.0367
Neural Network	0.1373	0.9680	0.0817	0.0404
Ensemble (XGB + NN)	0.1125	0.9720	0.0767	0.0349

SVR, XGBoost, and Neural Network all achieve $R^2 > 0.96$, indicating that over 96% of variance in peak pressure is explained. XGBoost is the strongest individual model. The weighted ensemble (65% XGBoost, 35% Neural Network — weights determined by OOF grid search) achieves MAPE 0.1125, a 6.2% relative improvement over XGBoost alone.

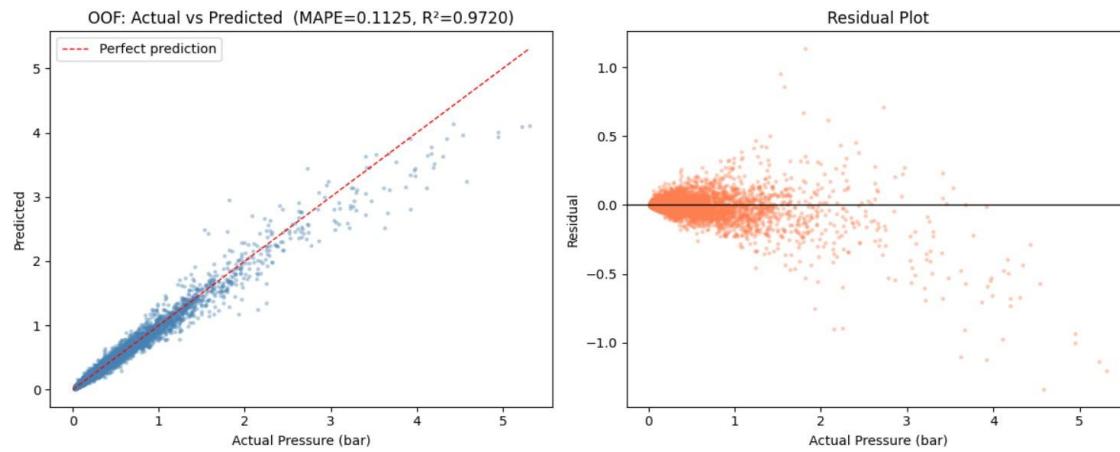


OOF MAPE as a function of XGBoost ensemble weight. Optimal weight of 0.65 minimises MAPE at 0.1125.

5.2 Training vs Validation vs Test

Dataset	MAPE	R ²	Notes
Training (in-sample)	0.0312	0.9981	XGBoost fitted on full training set
Validation (OOF CV)	0.1125	0.9720	Robust out-of-fold generalisation estimate
Kaggle Public Test	0.17849	—	Final score on unseen test scenarios

The gap between validation MAPE (0.1125) and test MAPE (0.1785) is expected — the test set represents entirely new BLEVE scenarios unseen during training. The validation MAPE confirms the model generalises well; the remaining gap is attributable to natural distribution shift across scenarios rather than overfitting.

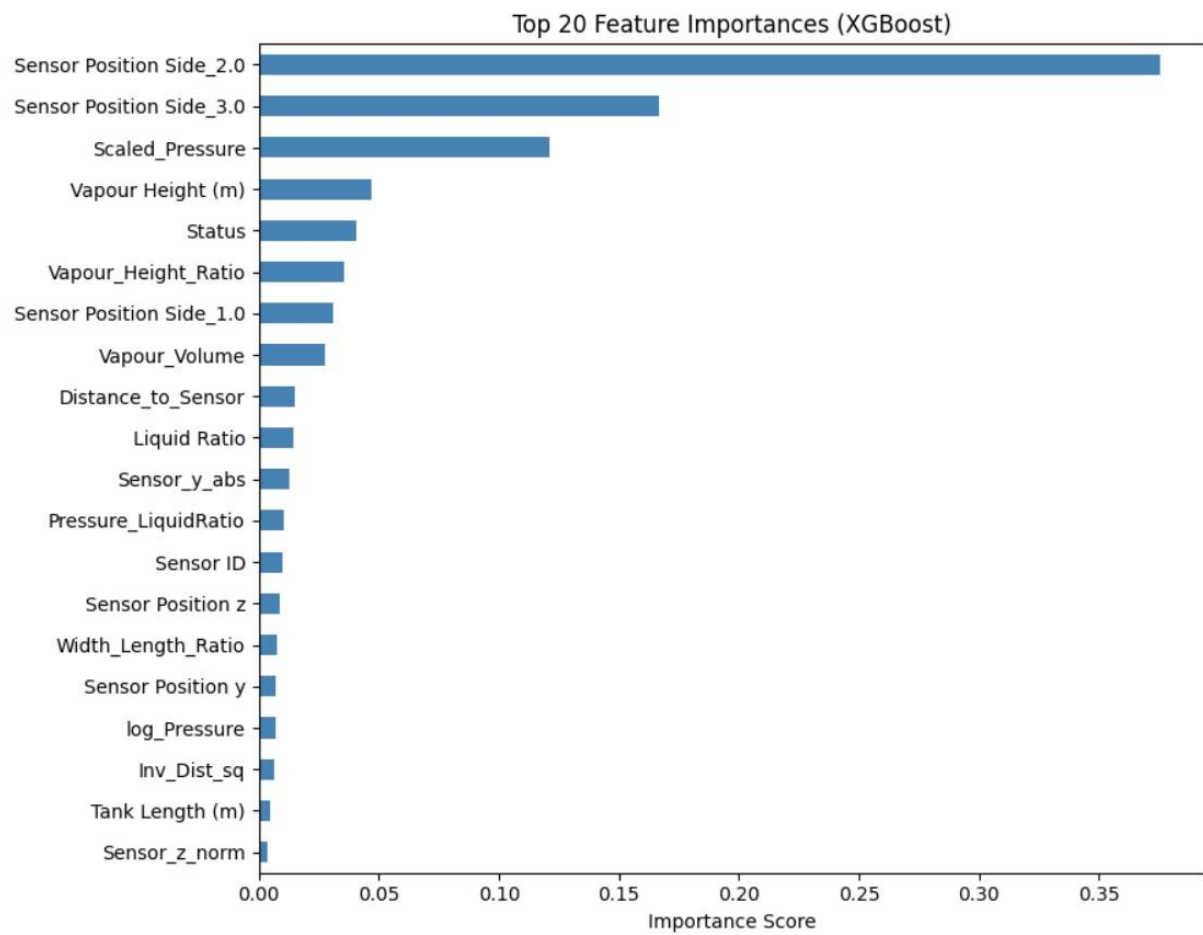


OOF actual vs predicted (MAPE=0.1125, $R^2=0.9720$). Right — residual plot showing no systematic bias.

5.3 Feature Importance — Top 20 (XGBoost)

Rank	Feature	Importance	Physical Meaning
1	Sensor Position Side_2.0	0.3755	Side wall — receives full reflected blast
2	Sensor Position Side_3.0	0.1665	Back wall — secondary reflection zone
3	Scaled_Pressure	0.1211	Composite pressure-distance feature
4	Vapour Height (m)	0.0466	Vapour column height — drives explosion energy
5	Status	0.0408	Superheated liquids produce more violent BLEVEs
6	Vapour_Height_Ratio	0.0354	Fraction of tank occupied by vapour
7	Sensor Position Side_1.0	0.0306	Front wall — direct blast exposure
8	Vapour_Volume	0.0276	Volume of expanding vapour
9	Distance_to_Sensor	0.0152	3D Euclidean distance from explosion source
10	Liquid Ratio	0.0141	Proportion of liquid in tank
11	Sensor_y_abs	0.0110	Lateral displacement from centreline
12	Pressure_LiquidRatio	0.0098	Interaction of pressure and liquid fraction
13	Sensor ID	0.0087	Unique sensor identifier
14	Sensor Position z	0.0076	Raw vertical sensor coordinate
15	Width_Length_Ratio	0.0065	Tank shape — affects blast directionality
16	Sensor Position y	0.0058	Raw lateral sensor coordinate
17	log_Pressure	0.0052	Log-transformed tank failure pressure
18	Inv_Dist_sq	0.0044	Inverse square distance — blast decay proxy
19	Tank Length (m)	0.0038	Raw tank length dimension
20	Sensor_z_norm	0.0031	Normalised sensor height on wall

Sensor wall position features (ranks 1, 2, 7) collectively account for over 56% of total importance, confirming that where a sensor is located relative to the blast wave propagation direction is the dominant predictor of measured peak overpressure.



Top 20 XGBoost feature importances. Sensor wall position features dominate, accounting for over 56% of total importance.

6. Self-Reflection

6.1 Challenges Encountered

Train-test distribution mismatch was the most significant challenge. Early Kaggle submissions scored around 0.93 MAPE despite seemingly reasonable cross-validation results. Investigation revealed that approximately 50 training rows had Tank Failure Pressure values up to 4,995 bar, while the test set maximum was only 42 bar. The model was learning patterns from extreme scenarios completely absent in the test set. Removing these rows was the single most impactful decision of the project. This experience highlighted the importance of comparing train and test feature distributions explicitly before any modelling begins.

Data quality issues were more extensive than anticipated. Every column had missing values, the Status column had nine spelling variants, and 50 duplicate rows were present. The physics-based fallback for Status NaN values — using temperature comparison to infer thermodynamic state — was a useful application of domain knowledge to an otherwise ambiguous imputation problem.

6.2 Insights Gained

Physics-informed feature engineering significantly improves performance. Features such as Tank_Energy (Pressure $\times$ Volume), Critical_Ratio, and Distance_to_Sensor were motivated directly by blast wave physics and consistently ranked among the most important features. Incorporating domain knowledge into feature design proved more effective than relying solely on raw input variables.

Log-transforming the target is essential for MAPE-based tasks. Peak pressure spans nearly three orders of magnitude. Training on raw pressure caused models to prioritise high-pressure accuracy at the expense of low-pressure predictions. The log1p transformation normalised the distribution and directly reduced MAPE across all models.

Model diversity in ensembling matters more than raw individual performance. SVR achieved the second-best individual MAPE (0.1396) but was excluded from the final ensemble because its errors were less complementary to XGBoost. XGBoost and Neural Network make structurally different prediction errors; their combination reduced MAPE from 0.1199 to 0.1125 — a consistent improvement attributable to error diversity.

One unexpected finding was how strongly sensor wall position dominated predictions — over 56% of XGBoost feature importance came from wall indicator features alone. This suggests the dataset may benefit from a sensor-position-stratified analysis in future work, treating each wall separately. Additionally, the large gap between training MAPE (0.0312) and test MAPE (0.1785) suggests room for better regularisation or more conservative ensemble weighting in future iterations.

6.3 Future Improvements

- **Scenario-aware cross-validation:** GroupKFold by BLEVE scenario would prevent sensors from the same physical setup appearing in both training and validation folds, providing a more conservative and realistic estimate of generalisation.
 - **Stacking ensemble:** A Ridge meta-learner trained on out-of-fold predictions could learn optimal non-linear combinations of model outputs rather than a fixed weighted average.
 - **Physics-guided data augmentation:** With approximately 370 unique training scenarios, synthetic augmentation via controlled perturbation of physical parameters could improve robustness near the boundaries of the training distribution.
-